

floatCSEP: An application to deploy and conduct reproducible, prospective earthquake forecasting experiments

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Summary

floatCSEP is a Python application that standardizes and orchestrates the workflow of earthquake forecasting experiments. Based on principles established by the Collaboratory for the Study of Earthquake Predictability (CSEP, <https://cseptest.org>), it enables reproducible, transparent, and reusable experiments to evaluate earthquake forecasts. floatCSEP builds on the existing pyCSEP toolkit for core evaluation routines and adds the functionality needed to deploy and conduct entire experiments, including catalog handling, forecast generation, evaluation, visualization, and reporting. Accompanying tutorials illustrate experiment use cases, which users can extend to incorporate new models, alternative evaluation metrics, or different regions and timeframes. Ultimately, the goal of floatCSEP is to support new official CSEP experiments and to encourage independent researchers to validate their own models.

Background

Earthquake forecasts are probabilistic statements about future earthquake occurrence ([Jordan et al., 2011](#)), used for informing building codes, emergency response planning, and risk reduction strategies. Because earthquake occurrence is driven by complex and highly non-linear processes (e.g., [Geller, 1997](#); [Y. Y. Kagan, 1994](#)), forecasts should be expressed and evaluated in a probabilistic framework designed to describe their fundamental uncertainties ([Y. Kagan & Jackson, 1994](#)). To assess their reliability, further challenges are the large time scales required to collect sufficient observations for evaluation (especially of large earthquakes) and the multiple subjective biases from any post-hoc adjustments in modeling or evaluation environments (e.g., [Schorlemmer & Gerstenberger, 2007](#)). To address such challenges, the Collaboratory for the Study of Earthquake Predictability (CSEP) was established to facilitate rigorous, prospective forecasting experiments where all forecasting models, data sources, evaluation metrics, and related software are defined prior to the evaluation time period (e.g., [Jordan, 2006](#); [Schorlemmer et al., 2018](#)). CSEP experiments were carried out in so-called Testing Centers, i.e., hardware and software infrastructure designed to ensure (i) controlled access, (ii) reproducible environments for automated forecast generation, and (iii) long-term archiving of input data, metadata and results ([Zecher et al., 2010](#)). With this framework, CSEP has successfully hosted and published

experiments across diverse geographic regions, such as California, New Zealand, Italy, Japan, and globally (Bayona et al., 2021, 2022, 2023; Eberhard et al., 2012; Field, 2007; Gerstenberger & Rhoades, 2010; Iturrieta et al., 2024; Nanjo et al., 2011; Strader et al., 2018; Taroni et al., 2014, 2018; Tsuruoka et al., 2012; Werner et al., 2010; Zechar et al., 2013). These efforts have substantially advanced scientific rigor and established community standards in earthquake forecasting research, thereby contributing to better forecasts and seismic hazard assessments (e.g., Michael & Werner, 2018; Schorlemmer et al., 2018).

Statement of Need

Despite their achievements, the original CSEP Testing Centers used centralized, rigid software architectures that tightly coupled data management to local hardware, limiting reuse, scalability, and community engagement (e.g., Savran, Bayona, et al., 2022; Schorlemmer et al., 2018; Zechar et al., 2010). As noted by Mizrahi et al. (2024), there is broad consensus in the earthquake forecasting community that transparency and reproducibility are essential in forecast testing. However, due to the complexity of Testing Centers, independent researchers often require advanced technical expertise to access, reproduce, and analyze CSEP experiments. To overcome these limitations, the Python package pyCSEP (Graham et al., 2024; Savran, Bayona, et al., 2022; Savran, Werner, et al., 2022) was developed to provide core forecast evaluation routines that can be directly integrated into modelers' workflows.

However, pyCSEP alone lacks key features required to deploy and conduct entire forecasting experiments, such as interfacing with external model source code, automating catalog access, managing data, standardizing workflow execution, and generating summary reports. That is, additional software is required that provides Testing Center capabilities while remaining decoupled from specific hosting hardware. To meet this need, we developed floatCSEP, which manages the entire experiment lifecycle, from model integration and initial deployment to the incremental updating of input data, forecasts, results, and reports as new observations become available. It is intended for earthquake forecast model developers, institutions that run CSEP-style forecasting experiments, and the broader statistical seismology community. To our knowledge, no existing software provides this complete end-to-end testing workflow; commonly used seismology tools instead address only individual steps, such as catalog queries (e.g., ObsPy) or forecast evaluation (e.g., pyCSEP).

Software Overview

The primary objective of floatCSEP is to provide a portable, automated, and reproducible Testing Center environment that runs on a standard workstation (subject to experiment-specific runtime and memory demands) and scales to HPC systems for computationally intensive experiments, such as time-dependent models producing daily forecasts. Experiments are defined through human-readable YAML (yaml.org) configuration files, which are processed through a simple command-line interface to ensure ease of use even for users without extensive computational expertise. This declarative approach simplifies the experiment setup, standardizes its workflow (Figure 1) and enhances its reproducibility.

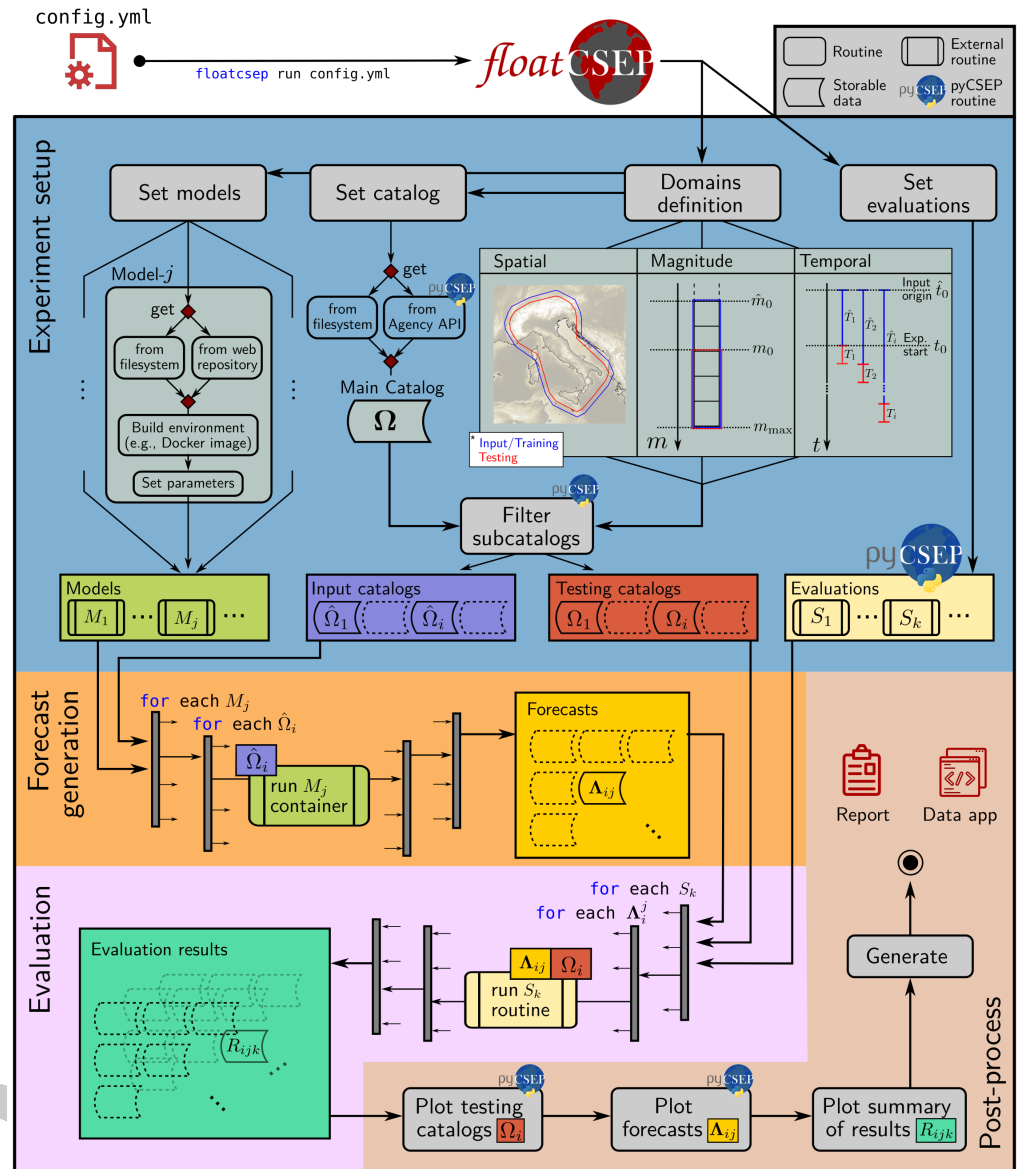


Figure 1: Workflow diagram of floatCSEP for a time-dependent experiment, which roughly consists of: 1) Defining time-space-magnitude ranges and discretizations for forecast generation and evaluation; 2) Querying and filtering earthquake catalogs (both for model input and evaluation); 3) Building the source code of external models, configuring its parameters and input data; 5) Generating forecasts by running each model source code in a containerized environment; 6) Performing forecast evaluations and comparisons using pyCSEP's or user-implemented testing metrics; and 7) Producing reports including test results and visual representations.

floatCSEP uses pyCSEP as a dependency, incorporating its core functionality (forecast and catalog classes, and evaluation routines) alongside additional Testing Center operations, such as data management and computational containerization (by using Docker, Merkel, 2014). The application supports multiple forecast formats and accommodates both time-invariant and time-variant experiments. It handles forecasts produced either by models managed directly by floatCSEP or provided externally through raw files. Representative use cases are included as tutorials, which users can extend by incorporating new models, adding alternative evaluations, or by replicating in different regions and timeframes. The software integrates seamlessly

89 with pyCSEP's existing testing routines and provides custom hooks for user-defined tests,
90 visualizations, and reports.

91 **Example Use**

92 An example configuration file (config.yml) for a time-invariant, grid-based experiment in Italy
93 with two models is shown in [Figure 2](#).



config.yml

```
name: CSEP Experiment                                * Experiment metadata
time_config:                                         * Temporal definition. Additional parameters
  start_date: 2010-1-1T00:00:00                    can create multiple windows
  end_date: 2020-1-1T00:00:00
region_config:                                       * Spatial and magnitude definitions (set jointly
  region: italy_csep_region                          as in pyCSEP format). region can refer to a
  mag_min: 3.95                                       CSEP existing region or to a file containing cell
  mag_max: 8.95                                       nodes
  mag_bin: 0.1
  depth_min: 0
  depth_max: 70
catalog: query_bsi                                  * Either a pyCSEP function or external file
models:
  - Smoothed-Seismicity:                            * Model's configuration. Can point to a file,
    path: ssm.dat                                     source code or web repository (e.g., github).
  - Spatially-Uniform:                               Model parameters are also set here
    path: uniform.dat
tests:
  - S-test:                                          * Tests to be carried out. Can point to
    func: poisson_evaluations.spatial_test            pyCSEP or custom functions.
    plot_func: plot_consistency_test
```

floatcsep {run, stage, plot, reproduce, view} config.yml

Figure 2: Simplest example of a configuration file for a time-invariant, grid-based experiment with two models. The run command executes the experiment end-to-end, generating and testing all missing forecasts from the last execution up to today. The stage command accesses and builds the models' source code, as well as preparing input and testing catalogs. The reproduce command re-executes an experiment and compares it with existing results using statistical and computational metrics. plot executes the post-process, image generation, and reporting of the experiment. view deploys a data app for visualizing the forecasts, catalogs, and test results.

94 **Applications**

95 floatCSEP is designed to support the following applications:

- 96 ■ Deploy and conduct new prospective experiments that incrementally incorporate new
97 data, update forecasts, and provide evaluation results. While the CSEP community
98 plans to use floatCSEP for new (official) experiments, we also encourage independent
99 researchers to adopt floatCSEP for prospectively evaluating their models
- 100 ■ Reproduce the results of completed prospective CSEP experiments within a containerized
101 computational environment (e.g., [Iturrieta et al., 2024](#)).
- 102 ■ Create new retrospective or pseudo-prospective experiments for their easy reproduction
103 and shareability.
- 104 ■ Plug in new models into a completed or ongoing (float)CSEP experiment. Since CSEP
105 experiments are clearly defined, they can be effectively used as benchmarks for comparing
106 and developing new forecasting models (e.g., [Serafini et al., 2025](#))

Support continuous evaluation of Operational Earthquake Forecasting systems that provide authoritative, near-real-time forecasts (e.g., Jordan et al., 2011; Mizrahi et al., 2024). However, most systems generate forecasts for overlapping windows (e.g., weekly forecasts updated daily) and evaluating the overall performance of such forecast collections remains an open methodological challenge (e.g., Brehmer et al., 2025).

floatCSEP contributes to a growing CSEP software ecosystem that, together with reproducibility packages (e.g., Allison et al., 2018; Bayona et al., 2022, 2023; Graham et al., 2024; Iturrieta et al., 2024; Savran, Bayona, et al., 2022), open-source forecasting models (e.g., Mizrahi et al., 2023), and long-term open-science repositories, could lay the foundation for building robust, collaborative benchmarks in earthquake forecasting.

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References

- Allison, D. B., Shiffrin, R. M., & Stodden, V. (2018). Reproducibility of research: Issues and proposed remedies. *Proceedings of the National Academy of Sciences*, 115(11), 2561–2562. <https://doi.org/10.1073/pnas.1802324115>
- Bayona, J. A., Savran, W. H., Iturrieta, P., Gerstenberger, M. C., Graham, K. M., Marzocchi, W., Schorlemmer, D., & Werner, M. J. (2023). Are regionally calibrated seismicity models more informative than global models? Insights from California, New Zealand, and Italy. *The Seismic Record*, 3(2), 86–95. <https://doi.org/10.1785/0320230006>
- Bayona, J. A., Savran, W. H., Rhoades, D. A., & Werner, M. J. (2022). Prospective evaluation of multiplicative hybrid earthquake forecasting models in California. *Geophysical Journal International*, 229(3), 1736–1753. <https://doi.org/10.1093/gji/ggac018>
- Bayona, J. A., Savran, W. H., Strader, A., Hainzl, S., Cotton, F., & Schorlemmer, D. (2021). Two global ensemble seismicity models obtained from the combination of interseismic strain measurements and earthquake-catalogue information. *Geophysical Journal International*, 224(3), 1945–1955. <https://doi.org/10.1093/gji/ggaa554>
- Brehmer, J. R., Kraus, K., Gneiting, T., Herrmann, M., & Marzocchi, W. (2025). Enhancing the statistical evaluation of earthquake forecasts—an application to Italy. *Seismological Research Letters*, 96(3), 1966–1988. <https://doi.org/10.1785/0220240209>
- Eberhard, D. A., Zechar, J. D., & Wiemer, S. (2012). A prospective earthquake forecast experiment in the western Pacific. *Geophysical Journal International*, 190(3), 1579–1592. <https://doi.org/10.1111/j.1365-246X.2012.05548.x>
- Field, E. H. (2007). Overview of the working group for the development of regional earthquake likelihood models (RELM). *Seismological Research Letters*, 78(1), 7–16. <https://doi.org/10.1785/gssrl.78.1.7>

- 152 Geller, R. J. (1997). Earthquake prediction: A critical review. *Geophysical Journal International*,
153 131(3), 425–450. <https://doi.org/10.1111/j.1365-246X.1997.tb06588.x>
- 154 Gerstenberger, M. C., & Rhoades, D. A. (2010). New zealand earthquake forecast test-
155 ing centre. *Pure and Applied Geophysics*, 167(8), 877–892. <https://doi.org/10.1007/s00024-010-0082-4>
- 156
- 157 Graham, K. M., Bayona, J. A., Khawaja, A. M., Iturrieta, P., Serafini, F., Biondini, E.,
158 Rhoades, D. A., Savran, W. H., Maechling, P. J., Gerstenberger, M. C., & others. (2024).
159 New features in the pyCSEP toolkit for earthquake forecast development and evaluation.
160 *Seismological Research Letters*, 95(6), 3449–3463. <https://doi.org/10.1785/0220240197>
- 161 Iturrieta, P., Bayona, J. A., Werner, M. J., Schorlemmer, D., Taroni, M., Falcone, G., Cotton,
162 F., Khawaja, A. M., Savran, W. H., & Marzocchi, W. (2024). Evaluation of a decade-long
163 prospective earthquake forecasting experiment in italy. *Seismological Research Letters*,
164 95(6), 3174–3191. <https://doi.org/10.1785/0220230247>
- 165 Jordan, T. H. (2006). Earthquake predictability, brick by brick. *Seismological Research Letters*,
166 77(1), 3–6. <https://doi.org/10.1785/gssrl.77.1.3>
- 167 Jordan, T. H., Chen, Y.-T., Gasparini, P., Madariaga, R., Main, I., Marzocchi, W., Papadopoulos,
168 G., Sobolev, G., Yamaoka, K., & Zschau, J. (2011). Operational earthquake forecasting:
169 State of knowledge and guidelines for utilization. *Annals of Geophysics*, 54(4), 315–391.
170 <https://doi.org/10.4401/ag-5350>
- 171 Kagan, Y. Y. (1994). Observational evidence for earthquakes as a nonlinear dynamic
172 process. *Physica D: Nonlinear Phenomena*, 77(1-3), 160–192. [https://doi.org/10.1016/0167-2789\(94\)90132-5](https://doi.org/10.1016/0167-2789(94)90132-5)
- 173
- 174 Kagan, Y., & Jackson, D. (1994). Long-term probabilistic forecasting of earthquakes. *Journal*
175 *of Geophysical Research: Solid Earth*, 99(B7), 13685–13700. <https://doi.org/10.1029/94JB00500>
- 176
- 177 Merkel, D. (2014). Docker: Lightweight linux containers for consistent development and
178 deployment. *Linux Journal*, 2014(239).
- 179 Michael, A. J., & Werner, M. J. (2018). Preface to the focus section on the collaboratory for the
180 study of earthquake predictability (CSEP): New results and future directions. *Seismological*
181 *Research Letters*, 89(4), 1226–1228. <https://doi.org/10.1785/0220180161>
- 182 Mizrahi, L., Dallo, I., Elst, N. J. van der, Christophersen, A., Spassiani, I., Werner, M. J.,
183 Iturrieta, P., Bayona, J. A., Iervolino, I., Schneider, M., & others. (2024). Developing,
184 testing, and communicating earthquake forecasts: Current practices and future directions.
185 *Reviews of Geophysics*, 62(3), e2023RG000823. <https://doi.org/10.1029/2023RG000823>
- 186 Mizrahi, L., Schmid, N., & Han, M. (2023). *Imizrahi/etas: ETAS with fit visualization* (Version
187 3.2). Zenodo. <https://doi.org/10.5281/zenodo.7584575>
- 188 Nanjo, K., Tsuruoka, H., Hirata, N., & Jordan, T. (2011). Overview of the first earthquake
189 forecast testing experiment in japan. *Earth, Planets and Space*, 63(3), 159–169. <https://doi.org/10.5047/eps.2010.10.003>
- 190
- 191 Savran, W. H., Bayona, J. A., Iturrieta, P., Asim, K. M., Bao, H., Bayliss, K., Herrmann, M.,
192 Schorlemmer, D., Maechling, P. J., & Werner, M. J. (2022). pyCSEP: A python toolkit
193 for earthquake forecast developers. *Seismological Society of America*, 93(5), 2858–2870.
194 <https://doi.org/10.1785/0220220033>
- 195 Savran, W. H., Werner, M. J., Schorlemmer, D., & Maechling, P. (2022). pyCSEP: A
196 python package for earthquake forecast developers. *Journal of Open Source Software*.
197 <https://doi.org/10.21105/joss.03658>
- 198 Schorlemmer, D., & Gerstenberger, M. (2007). RELM testing center. *Seismological Research*

- 199 *Letters*, 78(1), 30–36. <https://doi.org/10.1785/gssrl.78.1.30>
- 200 Schorlemmer, D., Werner, M. J., Marzocchi, W., Jordan, T. H., Ogata, Y., Jackson, D. D., Mak,
201 S., Rhoades, D. A., Gerstenberger, M. C., Hirata, N., & others. (2018). The collaboratory
202 for the study of earthquake predictability: Achievements and priorities. *Seismological*
203 *Research Letters*, 89(4), 1305–1313. <https://doi.org/10.1785/0220180053>
- 204 Serafini, F., Bayona, J. A., Silva, F., Savran, W., Stockman, S., Maechling, P. J., & Werner, M.
205 J. (2025). A benchmark database of ten years of prospective next-day earthquake forecasts
206 in california from the collaboratory for the study of earthquake predictability. *Scientific*
207 *Data*, 12(1), 1501. <https://doi.org/10.1038/s41597-025-05766-3>
- 208 Strader, A., Werner, M. J., Bayona, J. A., Maechling, P., Silva, F., Liukis, M., & Schorlemmer,
209 D. (2018). Prospective evaluation of global earthquake forecast models: 2 yrs of observa-
210 tions provide preliminary support for merging smoothed seismicity with geodetic strain rates.
211 *Seismological Research Letters*, 89(4), 1262–1271. <https://doi.org/10.1785/0220180051>
- 212 Taroni, M., Marzocchi, W., Schorlemmer, D., Werner, M. J., Wiemer, S., Zechar, J. D.,
213 Heiniger, L., & Euchner, F. (2018). Prospective CSEP evaluation of 1-day, 3-month, and
214 5-yr earthquake forecasts for italy. *Seismological Research Letters*, 89(4), 1251–1261.
215 <https://doi.org/10.1785/0220180031>
- 216 Taroni, M., Zechar, J., & Marzocchi, W. (2014). Assessing annual global m 6+ seismicity
217 forecasts. *Geophysical Journal International*, 196(1), 422–431. [https://doi.org/10.1093/](https://doi.org/10.1093/gji/ggt369)
218 [gji/ggt369](https://doi.org/10.1093/gji/ggt369)
- 219 Tsuruoka, H., Hirata, N., Schorlemmer, D., Euchner, F., Nanjo, K. Z., & Jordan, T. H. (2012).
220 CSEP testing center and the first results of the earthquake forecast testing experiment in
221 japan. *Earth, Planets and Space*, 64(8), 661–671. [https://doi.org/10.5047/eps.2012.06.](https://doi.org/10.5047/eps.2012.06.007)
222 [007](https://doi.org/10.5047/eps.2012.06.007)
- 223 Werner, M. J., Zechar, J. D., Marzocchi, W., & Wiemer, S. (2010). Retrospective evaluation
224 of the five-year and ten-year CSEP-italy earthquake forecasts. *Annals of Geophysics*, 53(3),
225 11–30. <https://doi.org/10.4401/ag-4840>
- 226 Zechar, J. D., Schorlemmer, D., Liukis, M., Yu, J., Euchner, F., Maechling, P. J., & Jordan,
227 T. H. (2010). The collaboratory for the study of earthquake predictability perspective on
228 computational earthquake science. *Concurrency and Computation: Practice and Experience*,
229 22(12), 1836–1847. <https://doi.org/10.1002/cpe.1519>
- 230 Zechar, J. D., Schorlemmer, D., Werner, M. J., Gerstenberger, M. C., Rhoades, D. A., & Jordan,
231 T. H. (2013). Regional earthquake likelihood models i: First-order results. *Bulletin of the*
232 *Seismological Society of America*, 103(2A), 787–798. <https://doi.org/10.1785/0120120186>